



Database to Actionable Insights

Query-Ecommerce

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Business Story

- Query ECommerce → Trouble with turning data into reliable and actionable insights
 - a. Unorganized data
 - b. Two different datasets
 - c. Missing Data → misleading data

We combined `dwsql_course` transaction data with demographic information (population, median income, region, urbanization).

- The goal
 - a. Provide clear & actionable insights
 - b. Optimizing marketing, resource allocation, and strategic investments

Business Question

Which states generate the highest per-capita sales, and what demographic characteristics explain these differences?

- **Population, Income, Region, Urban percentage**

Performed 6 types of analysis

- 1. Total Revenue by State**
- 2. Per-Capita Sales Ranking**
- 3. Revenue Trends Across Population Segments**
- 4. Three-Month Rolling Average Sales Trend**
- 5. Regional Revenue and AOV Comparison**
- 6. Income-Tier Spending Analysis**

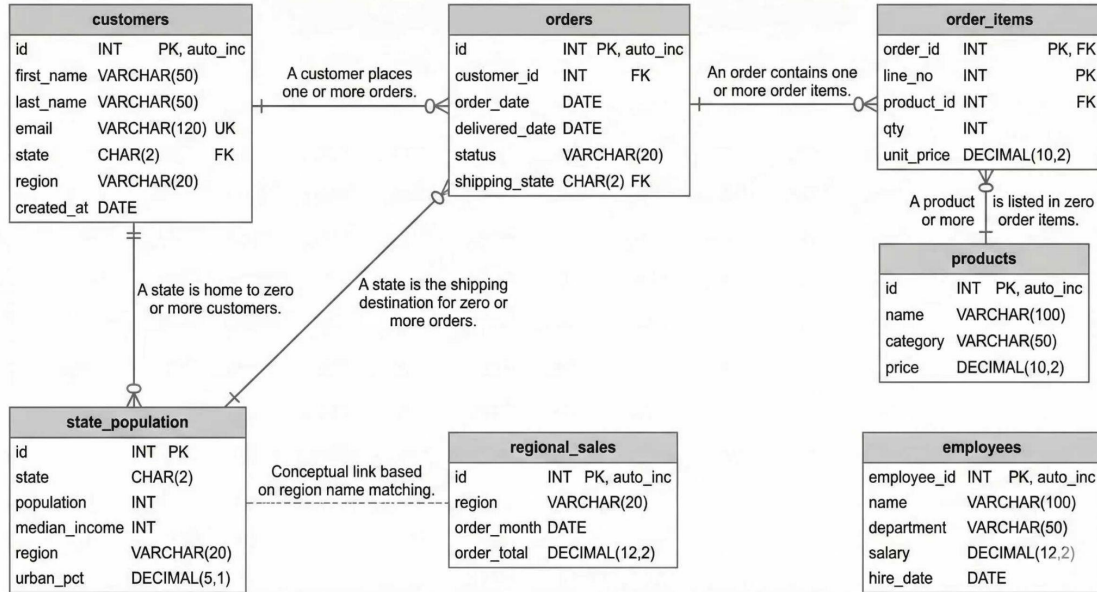
Query Setup

Entities	Attributes
Customers	id, first_name, last_name, email, state, region, created_at
Employees	empolyee_id, name, department, salary, hire_date
Order_Item	order_id, line_no, product_id, qty, unit_price
Orders	id, customer_id, order_date, delivered_date, status, shipping_state
Products	id, name, category, price
Regional_Sales	id, region, order_month, order_total
State_Population	state, population, median_income, region, urban_pct

Entity Relationship

Orders → Customers	(M:1)
Order_Item → Orders	(M:1)
Customers → State_Population	(M:1)
Orders → State_Population	(M:1)
Order_Item → Products	(M:1)
State_Population → Regional_Sales	(M:1)

Relationships



1. A customer can place multiple orders, but an order is linked to exactly one customer.
2. An order can contain multiple order items, but an order item belongs to exactly one order.
3. A product can appear in multiple order items, but an order item refers to exactly one product.
4. A customer has one home state, but a state can have multiple customers.
5. An order is shipped to one state, but a state can receive multiple orders.
6. A state_population row describes one state, but a state can be related to multiple customers and orders through that state code.
7. A state is assigned to one region in the state_population table, but a region can include multiple states.
8. A regional_sales record summarizes one region in one month, but a region can have multiple regional_sales records across months.

Finding: Sales + Three-Month Moving Average

Figure 1:

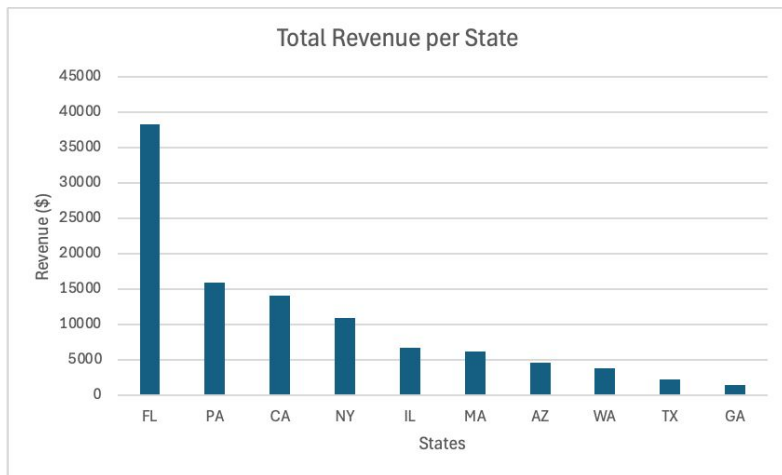


Figure 1. Bar graph of total sales per state (Joining order_items and orders on order_id)

Figure 2:

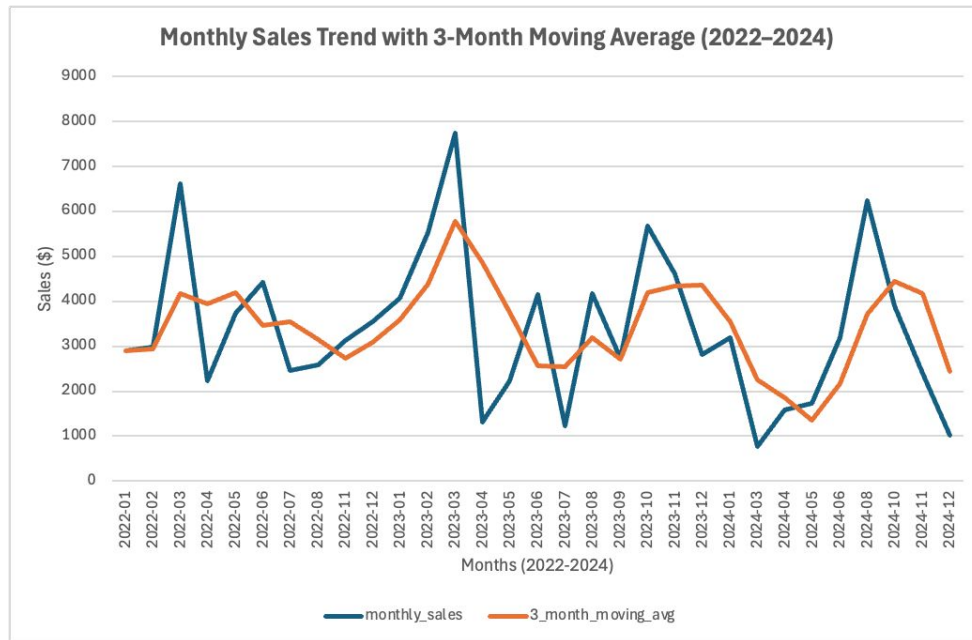


Figure 2: Two-series line chart of the 3 month moving average vs. monthly sales (Windows function average)

Finding: Sales vs. State

```

WITH StateSales AS (
    -- CTE: Calculates total revenue per state
    SELECT
        o.shipping_state,
        SUM(oi.qty * oi.unit_price) as state_revenue
    FROM orders o
    JOIN order_items oi ON o.id = oi.order_id
    GROUP BY o.shipping_state
)
SELECT
    s.shipping_state,
    p.region,
    s.state_revenue,
    p.population,
    -- Calculate sales per capita
    ROUND((s.state_revenue / p.population), 4) as sales_per_capita,
    -- Window Function: Ranks states based on per-capita performance
    DENSE_RANK() OVER (ORDER BY (s.state_revenue / p.population) DESC) as per_capita_rank
FROM StateSales s
JOIN state_population p ON s.shipping_state = p.state
ORDER BY per_capita_rank;
    
```

Alias for temporary results

Aggregate Function

Joins

Grouping

Final Output Columns

Derived Column

Rank Rows

Sort Order

shipping_state	region	state_revenue	population	sales_per_capita	per_capita_ra...
FL	South	38327.04	21538187	0.0018	1
PA	Northeast	15921.61	13002700	0.0012	2
MA	Northeast	6272.18	7029917	0.0009	3
AZ	West	4712.73	7276316	0.0006	4
NY	Northeast	11016.00	19835913	0.0006	5
IL	Midwest	6715.46	12812508	0.0005	6
WA	West	3879.62	7705281	0.0005	7
CA	West	14161.88	39538223	0.0004	8
GA	South	1506.90	10711908	0.0001	9
TX	South	2329.51	29145505	0.0001	10

Figure 3:

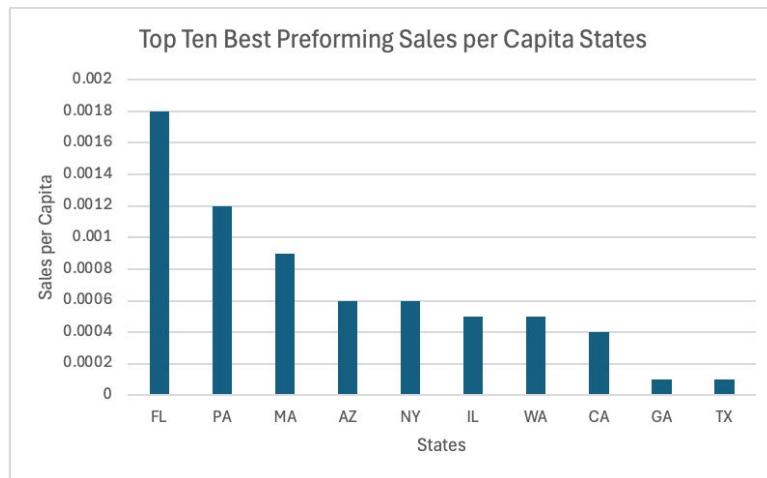


Figure 3: Bar Chart of Sales per capita of each state (Use CTE create temporary table StateSales, joining revenue data with population data, and window functions for ranking)

Finding: Population vs. Revenue

Figure 4:

population_seg...	state_cou...	avg_revenue_per_state	avg_urban_pct	
High Pop (>10M)	7	12854.06	86.0	
Med Pop (5M-10M)	3	4954.84	88.6	

Figure 4: Hypothesis Correlation, Revenue vs. Population (Group states by population) and the average urban percentage per population segment.

Population Segments:

1. High population: Greater than 10 million (>10M)
2. Medium Population: Between 5 million and 10 million (5M-10M)
3. Small Population: Less than 5 million (<5M)

```
SELECT
    CASE
        WHEN p.population > 10000000 THEN 'High Pop (>10M)'
        WHEN p.population > 5000000 THEN 'Med Pop (5M-10M)'
        ELSE 'Small Pop (<5M)'
    END AS population_segment,
    COUNT(DISTINCT s.shipping_state) AS state_count,
    ROUND(AVG(s.state_revenue), 2) AS avg_revenue_per_state,
    ROUND(AVG(p.urban_pct), 1) AS avg_urban_pct
```

Annotations in the image:

- A bracket on the left side of the CASE statement is labeled "Case expression for categorization".
- An arrow points from the text "New field" to the AS population_segment, line.

Finding: Regional Sales

region	region_total_sales	region_population	region_per_capita
South	42163.45	126296072	0.00033
Northeast	33209.79	57243812	0.00058
West	22754.23	78713578	0.00029
Midwest	6715.46	68985454	0.00010

Figure 5:

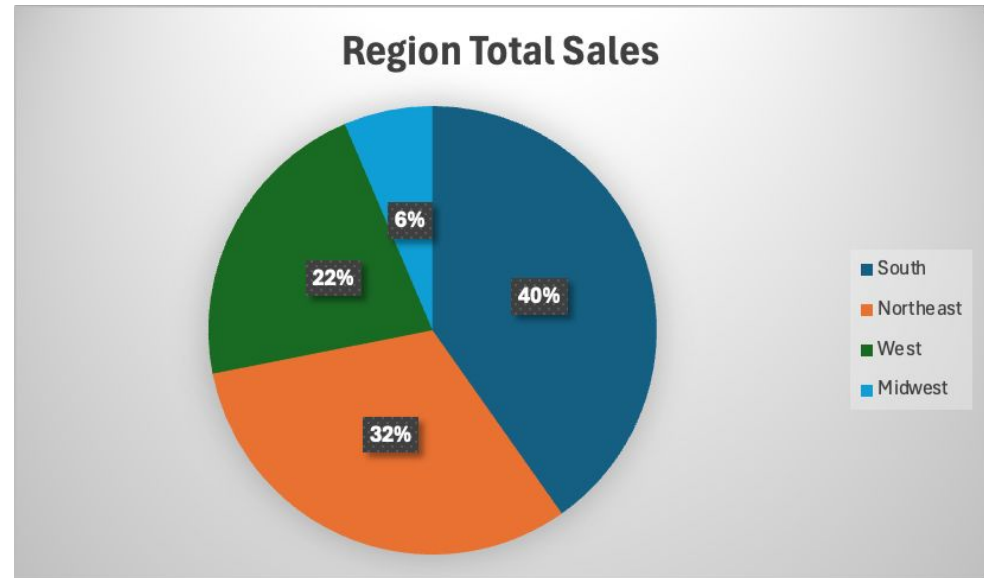


Figure 5: Pie Chart of region_total_sales from 'v_final_state_metrics'

Recommendation

- Prioritize high per-capita states like FL and PA and low per-capita states
- Use demographic segmentation to target
 - High-population
 - Urban states
- Strengthen regional strategy in the South and Northeast because they contribute the largest share of total revenue.
- We want to maintain consistent performance in high performing states and regions while also find way to improve sales on low performing areas

Future Steps

- Integrate additional external variables for deeper insight.**
- Build predictive models to forecast high-value states.**
- Run A/B experiments to validate marketing strategies.**
- Expand our data to reduce standard error.**

An aerial photograph of Central Park in New York City, showing the Bethesda Fountain in the foreground, the Bow Bridge in the middle ground, and the dense Manhattan skyline in the background. The image is overlaid with a semi-transparent purple banner at the top and white text in the center. A purple quotation mark is positioned above the text, and a purple horizontal line is positioned below it.

“
THANK YOU FOR
YOUR TIME!
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